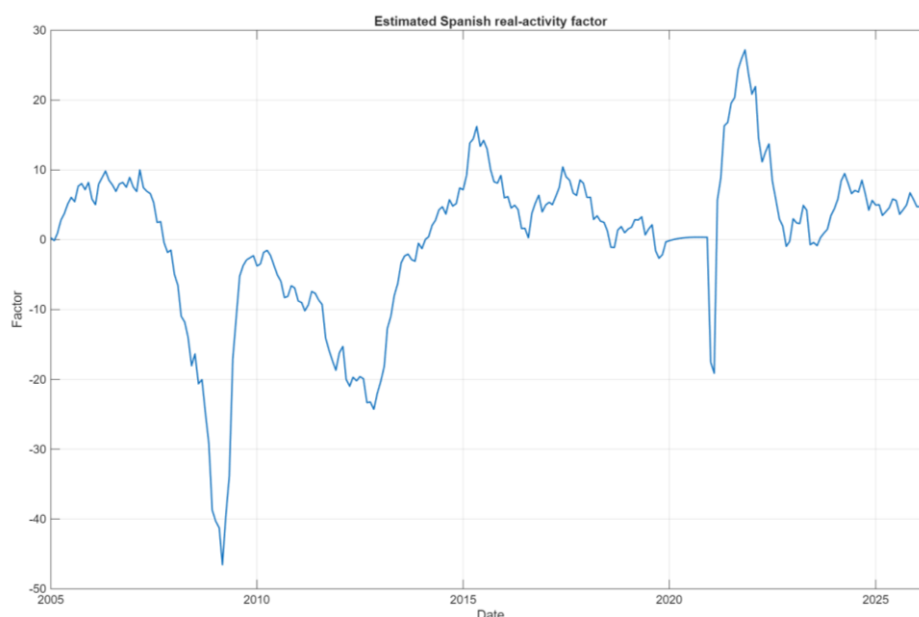
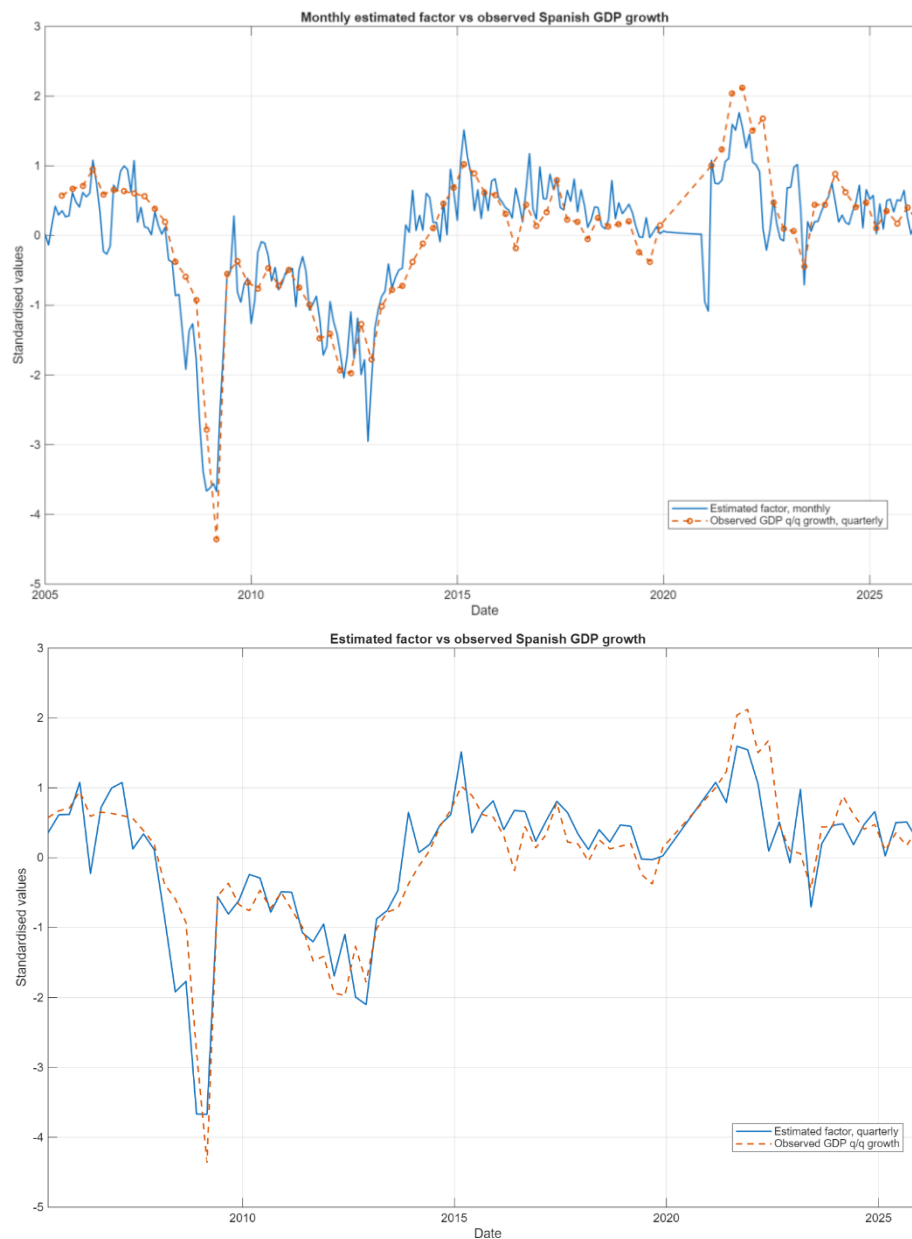


1. The country selected is Spain.
2. I use one quarterly target variable and six monthly indicators, one of which is a soft indicator.
3. The variables I chose are:
 - a. Quarterly real GDP growth, which is the target variable. (Eurostat)
 - b. The industrial production, as industrial activity and production is a classical key element to study the state of the economy. Furthermore, Spain presents a solid industrial sector, so industrial production can be a useful proxy for real economic activity. (Eurostat)
 - c. Social Security affiliation, to proxy the labor market and demand. Spain has historically suffered from high unemployment, so labour-market conditions are especially relevant for assessing its economic cycle. (Seguridad Social)
 - d. Retail sales, to proxy more closely the private consumption, also a classic standard variable. (Eurostat)
 - e. Imports-related external payments, as a monthly proxy for domestic demand and production needs, as stronger internal activity tends to increase payments to the rest of the world for goods, services, intermediate inputs and capital goods. (ECB)
 - f. Tourism overnights, as a proxy for tourism, as tourism is a key component of Spanish economic activity, representing 12.6% of the GDP in 2024 (INE). (Eurostat)
 - g. As a soft indicator, I chose the Economic Sentiment Indicator, to proxy the economic sentiment in Spain. (Eurostat)

I chose 2005 as the start date to have enough time sample to have a solid model but not suffer from lacking most variables during early times, nor forcing the model to adapt to different regimes with the same intrinsic relationship.

4. The following figure is the plot of the estimated factor.





In the previous two figures we can observe how the estimated factor tracks the real GDP growth. The first one shows the monthly estimated factor, which already shows close track of economic activity, although it is noisy. That is why we use the quarterly comparison as a second figure, to better see how they both are closely related. To assess its relevance compared with the GDP it was measured that the correlation between them is 0.899.

5. Quarter Forecast q/q GDP growth

2026Q2 0.7644

2026Q3 0.7677